Case Study 2 - William Armentrout

Part 1: Population, Sample, and Unit Analysis

The original GSS dataset population aims to capture a random sample of adults from the United States. One unit of analysis in this population survey is supposed to represent one respondent to the survey. Each row in the survey aims to accurately represent one individual adult in the United States. The sample of the survey used in this case study is a 1,500-row subset of the full GSS, put into a file called GSS_edited_values.xlsx. The sampling frame for the edited file is unknown as the file does not include the full sampling frame. This means we can't know some information about the survey such as year taken, survey weights, stratification, clustering, or design of data collection. Because we don't have the sampling method of the full GSS dataset we should not assume that the 1,500 subset of the GSS accurately measures a random sample of all U.S. adults. This means conclusions like, 8.53% of adults in the US are between the ages of 18 and 24, would be inappropriate to make from this edited subset.

Part 2: Sample Distribution

Below are frequency tables created with the use of a frequency helper function that will find the count and percentage of any provided variable from the GSS file. For the rincome91 table an additional mapping of the original data is used to merge groups together to limit the size of the table and make it easier to read while maintaining the usefulness of the frequency table. The mapping also merged responses such as "REFUSED", "DK", and "NA" all as "Non-substantive" as they held similar information. "NAP" was kept as for rincom91 it means the respondent wasn't asked the question.

age_group	count	percent
18–24	128.0	8.53
25–34	304.0	20.27
35–44	368.0	24.53
45–54	243.0	16.20
55–64	167.0	11.13
65–74	163.0	10.87
75+	122.0	8.13
No answer	5.0	0.33
Total	1500.0	100.00

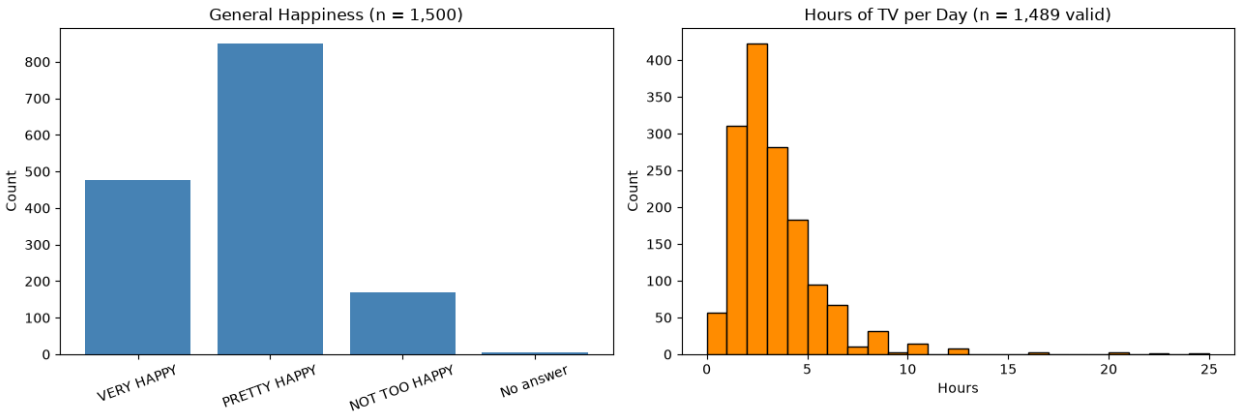
sex	count	percent
Female	859.0	57.27
Male	641.0	42.73
Total	1500.0	100.00

marital	count	percent
MARRIED	795.0	53.00
NEVER MARRIED	286.0	19.07
DIVORCED	213.0	14.20
WIDOWED	165.0	11.00
SEPARATED	40.0	2.67
NA	1.0	0.07
TOTAL	1500.0	100.00

happy	count	percent
VERY HAPPY	477.0	31.80
PRETTY HAPPY	849.0	56.60
NOT TOO HAPPY	169.0	11.27
No answer	5.0	0.33
Total	1500.0	100.00

rincome91	count	percent
Under $10K	208.0	13.87
$10,000-19,999	231.0	15.40
$20,000-29,999	200.0	13.33
$30,000-39,999	137.0	9.13
$40,000-59,999	104.0	6.93
$60,000+	67.0	4.47
NAP	493.0	32.87
Non-substantive	60.0	4.00
Total	1500.0	100.00

The bar chart and histogram below were generated using an age_group variable that uses the same grouping as the age frequency table above. This was then used in a python program using matplotlib to generate the charts.



The groups that are most represented in the edited sample are: Female (57.27%), MARRIED (53.00%), and ages 35-44 (24.53). The groups least represented are: SEPARATED (2.67%), age 75+ (8.13%) and 18-24 (8.53%). In terms of skewness and unevenness, happy heavily skews towards PRETTY HAPPY (56.6%), rincom91 has a rather large NAP block (32.87%).

This 1,500 record sample does not guarantee an exact representation of every subgroup because as mentioned above in part 1 we don't have the full sampling frame or data collection method for the full data. Even if we did know the full sampling frame or data collection we still couldn't guarantee subgroup representation as sampling variation means the proportions can differ from the true population especially for small subgroups. What we can infer is that the relative proportions within this dataset are accurate, but we can't infer anything about the population level estimates.

Part 3: Missingness and Non-Substantive Responses

Step 1: Define Messingness

For this study there was some messiness in the fields specifically when it comes to absent values. To handle this we grouped all values such as DK, no answer, refused, blank, and system-missing as "non-substantive". But we did decide that we would leave NAP or not applicable as a separate value as this is not a refusal to answer the question, but rather a statement the question wasn't relevant to the person. For example, someone who is retired would not have an income.

Step 2: Create a Missingness Table

Variable	Valid substantive responses	Not applicable	Don't know / refused / no answer/ blank	System missing	Total records	Non-substantive missing %
Income (rincome91	947	493	60	0	1,500	4.00

Job satisfaction (satjob)	1143	309	48	0	1,500	3.20
General happiness (happy)	1495	0	5	0	1,500	0.33
Life after death (postlife)	906	485	109	0	1,500	7.27
Television hours (tvhours)	1489	0	11	0	1,500	0.73

Television hours contained 11 records with the missing values, when working with descriptive statistics it is a better practice to remove those 11 from the total 1,500 than to substitute in the value 0 hours. This is because in the 1,500 samples there are already some who reported 0 hours and to treat any missing values as 0 would skew the data more. It is also important to check for implausible values. Television hours is the hours of TV someone watches in a day having a possible range of 0 to 24. So we run a check for values exceeding 16 and 18 hours a day in tvhours and flag them as implausible. In the case of this sample there are 3 entries that are flagged by this check out of 1,500.

Step 3: Compare Missingness Across Groups

To look further into the messingness across groups we took a look into missing values across various age groups as shown in the table below. This table was generated using a missing_by_group helper function (similar to the frequency helper). The table includes two missing % values. The "of all" variant is meant to show how much of the data is missing and the "excel. N/A" shows among those who had a chance, how many didn't answer.

age_group	Total	Not applicable	Missing	Missing %(of all)	Missing % (excel. N/A)
18-24	128	46	1	0.78	1.22
25-34	304	53	8	2.63	3.19
35-44	368	55	15	4.08	4.79
45-54	243	42	23	9.47	11.44
55-64	167	60	2	1.20	1.87
65-74	163	130	3	1.84	9.09

75+	122	105	6	4.92	35.29
No answer	5	2	2	40.00	66.67

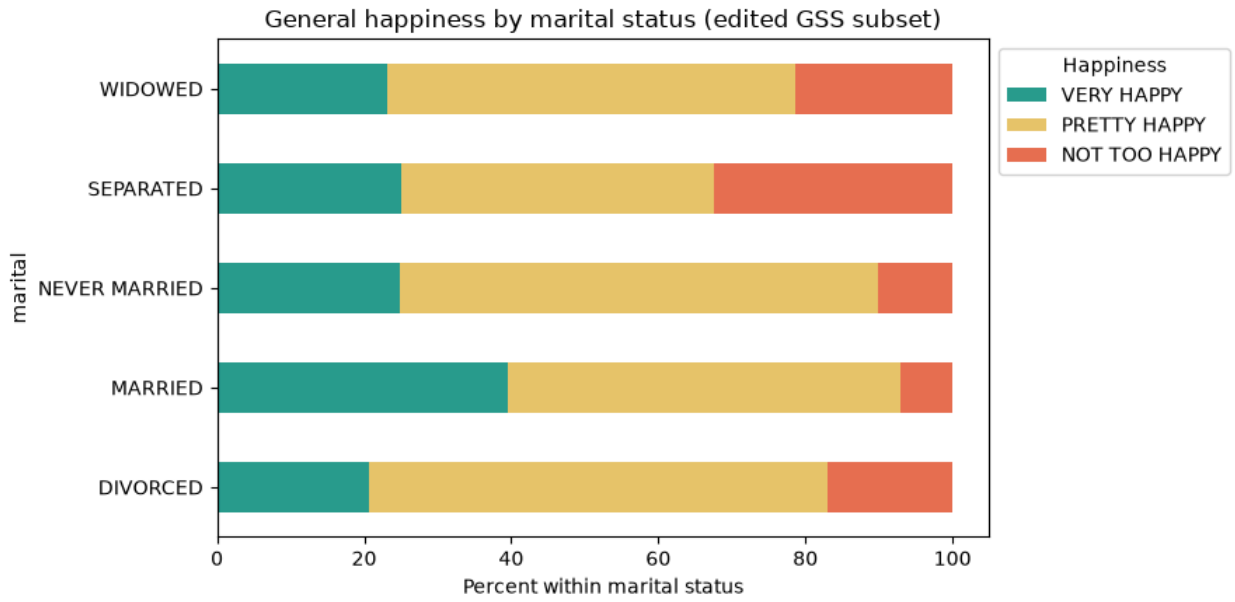
Step 4: Interpretation

The variables with the highest non-substantive missingness are postlife (7.27%) and income (4.00%). The variables with the lowest non-substantive missingness are happy (0.33%) and tvhours (0.73%). In the group comparison of missingness we can see that missingness is not evenly distributed within the groups. This is best seen with the 18-24 age group having a missingness of 1.22% and the 45-54 age group having 11.44%. This however does not establish that missingness is random within the dataset because there is no information on who declined or why. One way listwise deletion could bias results is that by dropping incomplete records would disproportionately remove more from certain age groups causing further skewing towards other groups.

Part 4: Sampling Bias and Measurement Bias Assessment

The lack of coverage and sampling-frame is a massive limitation as without that information it makes it impossible for us to assess how representative the survey is of the whole population. The messingness results from part 3 above indicate that people who decline to answer income or job questions differ on a number of factors. One example of this is in age groups, specifically with the youngest age group, 18-24, having a missingness value of 1.22% and the oldest age group, 75+, having a missingness value of 35.29%. There is also an example of response bias in the happiness value. People may be reporting that they are happier or sadder than they actually are.

Part 5: How does general happiness vary across marital-status?



The figure above was generated using matplotlib tools and the pandas crosstab function across the subset of 1,500 rows from the class dataset.

The question we wanted to explore was, How does general happiness vary across marital status in this dataset? We found that MARRIED has the highest VERY HAPPY share (~39%) while DIVORCED has the lowest. SEPARATED has the largest share of the NOT TOO HAPPY share. Keep in mind these are not causal relationships and may only indicate an association between the variables. Missingness was handled with non-substantive happy or marital values being dropped before the cross tabulation, keeping a total of 1,494 out of the 1,500. One limitation of this is the unequal number of responses for the categorical variables. The primary example being SEPARATED only has 40 individuals while MARRIED has 793 individuals. Another limitation is that happiness is a self reported variable, meaning that the reported value may not be representative of the actual value for each individual. Any conclusion drawn from this research question is important to note that it is a conclusion on the edited subset of 1,500 rows and cannot be extrapolated back to the original GSS dataset or back to the entire population.

Gen-AI usage:

Claude Sonnet 5 was used on Sept 28 and 30, 2026 for creating and debugging python code in a jupyter notebook file using the libraries pandas, numpy, and matplotlib for tables, frequency groupings and charts based on the GSS_editied_values.xlsx file. Prompts usually followed this style: “Create python code / function for a jupyter notebook cell to create a table/chart containing relevant information about the figure from the GSS file”.